# BREAST CANCER DETECTION

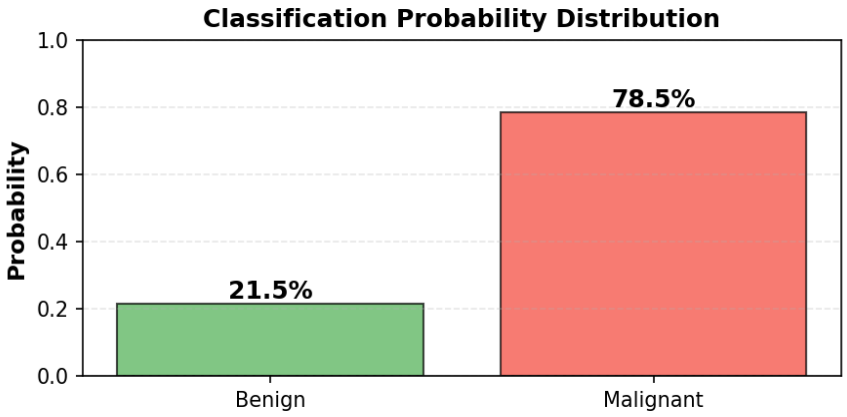
***AI-Assisted Histopathology Analysis Report***

| | |
|---|---|
| **Report ID:** | BCR-20260216_193507 |
| **Generated:** | 2026-02-16 19:35:08 |
| **Patient ID:** | N/A |
| **Image File:** | tmp2kodm6yh.png |

## DIAGNOSTIC RESULTS

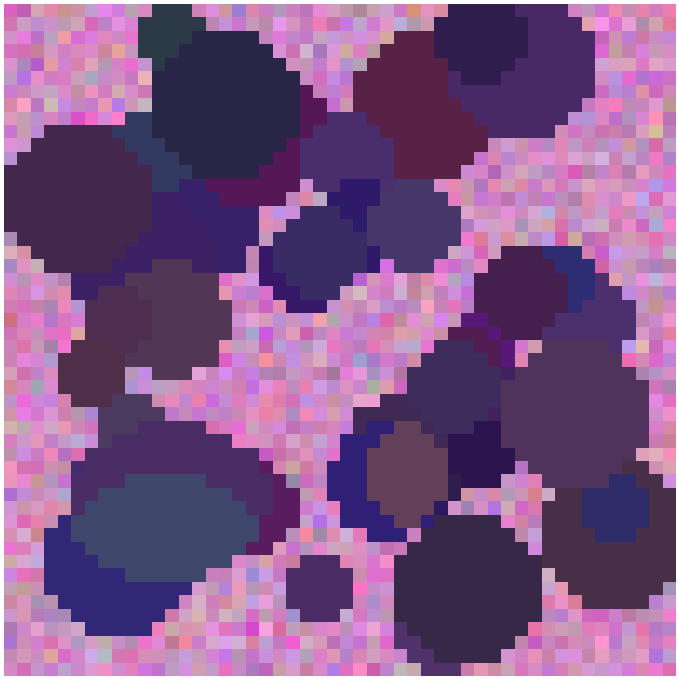
| | |
|---|---|
| **Classification:** | **MALIGNANT** |
| **Confidence Score:** | 78.5% |
| **Benign Probability:** | 21.5% |
| **Malignant Probability:** | 78.5% |

### Probability Distribution



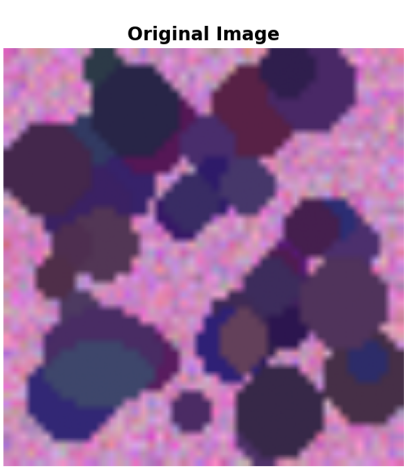

## HISTOPATHOLOGY IMAGE ANALYSIS

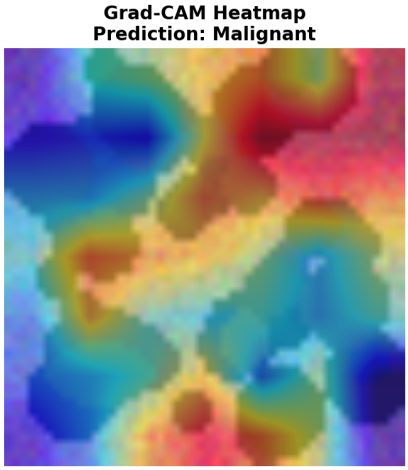
### Original Histopathology Image



## Explainability: Grad-CAM Heatmap

The heatmap below highlights the tissue regions that most influenced the AI's decision. Red/yellow areas indicate high importance, while blue areas indicate low importance.



## CLINICAL RECOMMENDATIONS

■■ RETEST AND SPECIALIST REVIEW RECOMMENDED. Moderate confidence malignant detection requires verification. Schedule immediate appointment with specialist for comprehensive evaluation. Additional imaging, biopsy, or pathological examination strongly advised to confirm findings and rule out false positive.

| | |
|---|---|
| **Risk Level:** | **HIGH** |
| **Follow-up Timeline:** | Immediate (within 24-48 hours) |
| **Specialist Referral:** | Oncologist and Pathologist |

## IMPORTANT MEDICAL DISCLAIMER

**This report is generated by an AI-assisted diagnostic system and is intended for decision support purposes only.**

**Key Points:**
• This analysis must be validated by qualified medical professionals
• Do NOT make treatment decisions based solely on this AI analysis
• Always consult with board-certified pathologists and oncologists
• This system is not a substitute for professional medical judgment
• False positives and false negatives can occur
• Clinical correlation and additional diagnostic tests are essential

**Regulatory Status:**
This AI system is for research and educational purposes. It has not been approved by regulatory agencies (FDA, CE, etc.) for clinical diagnostic use.

**Professional Review Required:**
All findings must be confirmed through standard pathological examination and clinical evaluation by licensed healthcare providers.

## TECHNICAL DETAILS

**Model Architecture:** ResNet50 with Transfer Learning
**Training Dataset:** Histopathology images (Benign/Malignant classification)
**Input Image Size:** 224 × 224 pixels
**Preprocessing:** ImageNet normalization
**Explainability Methods:** Grad-CAM, SHAP, Integrated Gradients
**Software Version:** Breast Cancer Detection System v1.0
**Processing Timestamp:** 2026-02-16 19:35:08

*This report was automatically generated by the AI-Assisted Breast Cancer Detection System. For questions or concerns, please consult with your healthcare provider.*